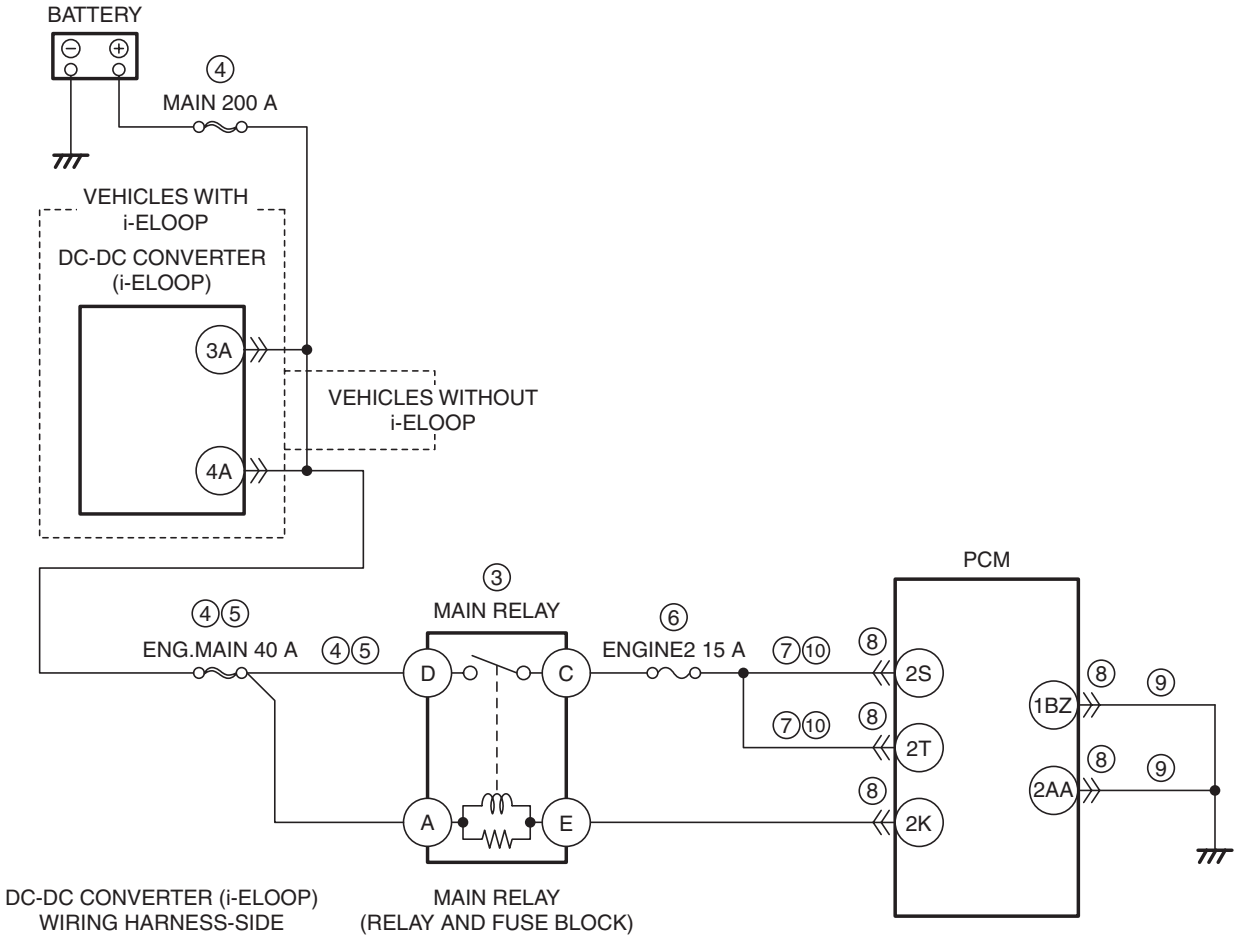
DTC P0685:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0102q1302800

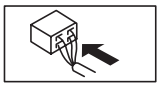
DTC P0685:00	Main relay control circuit open
DETECTION CONDITION	• The period of time in which the PCM power supply remains on after the ignition is switched off is not within the specified time. **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition in two consecutive drive cycles or in one drive cycle while the DTC for the same malfunction has been stored in the PCM. • PENDING CODE is available if the PCM detects the above malfunction condition during first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• Main relay malfunction • Short to ground or open circuit in main relay power supply circuit: (without i-ELOOP) — Short to ground in wiring harness between MAIN 200 A fuse and main relay terminal D — MAIN 200 A fuse and/or ENG.MAIN 40 A fuse malfunction — Open circuit in wiring harness between battery positive terminal and main relay terminal D • Short to ground or open circuit in main relay power supply circuit: (with i-ELOOP) — Short to ground in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and main relay terminal D — ENG.MAIN 40 A fuse malfunction — Open circuit in wiring harness between DC-DC converter (i-ELOOP) terminal 4A and main relay terminal D • ENGINE2 15 A fuse malfunction • Short to ground in wiring harness between the following terminals: — Main relay terminal C—PCM terminal 2S — Main relay terminal C—PCM terminal 2T • PCM connector or terminals malfunction • Open circuit in wiring harness between PCM terminal 1BZ and body ground • Open circuit in wiring harness between PCM terminal 2AA and body ground • Open circuit in wiring harness between the following terminals: — Main relay terminal C—PCM terminal 2S — Main relay terminal C—PCM terminal 2T • PCM malfunction

DTC P0685:00	Main relay control circuit open

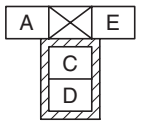


DC-DC CONVERTER (i-ELOOP) WIRING HARNESS-SIDE CONNECTOR

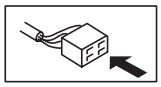




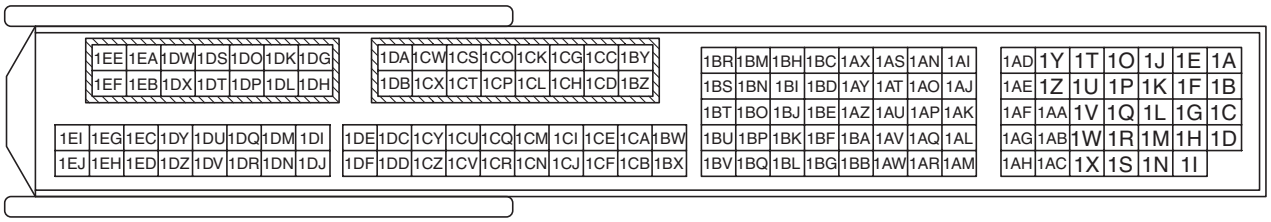
MAIN RELAY (RELAY AND FUSE BLOCK)

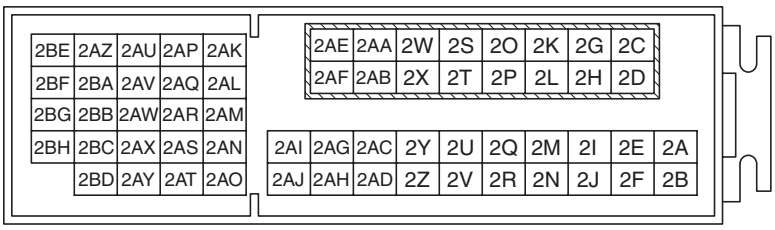


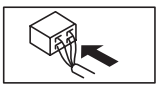
FRONT



PCM WIRING HARNESS-SIDE CONNECTOR







Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT MAIN RELAY** • Switch the ignition off. • Remove the main relay. (See RELAY LOCATION.) • Inspect the main relay. (See RELAY INSPECTION.) • Is there any malfunction?	Yes	Replace the main relay, then go to Step 11.
		No	Vehicles with i-ELOOP: • Go to Step 5. Vehicles without i-ELOOP: • Go to the next step.
4	**INSPECT MAIN RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Main relay is removed. • Measure the voltage at the main relay terminal D (wiring harness-side). • Is the voltage **B+**?	Yes	Go to Step 6.
		No	Inspect the MAIN 200 A fuse and ENG.MAIN 40 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between MAIN 200 A fuse and main relay terminal D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. • Replace the malfunctioning fuse. • If the fuse is damaged: — Replace the malfunctioning fuse. • If all fuses are normal: — Refer to the wiring diagram and verify whether or not there is a common connector between battery positive terminal and main relay terminal D. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 11.

STEP	INSPECTION		ACTION
5	**INSPECT MAIN RELAY POWER SUPPLY CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Main relay is removed. • Measure the voltage at the main relay terminal D (wiring harness-side). • Is the voltage **B+**?	Yes	Go to the next step.
		No	Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Inspect the ENG.MAIN 40 A fuse. • If the fuse is blown: — Refer to the wiring diagram and verify whether or not there is a common connector between ENG.MAIN 40 A fuse and main relay terminal D. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has shorted to ground, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. • Replace the fuse. • If the fuse is damaged: — Replace the fuse. • If the fuse is normal: — Refer to the wiring diagram and verify whether or not there is a common connector between DC-DC converter (i-ELOOP) terminal 4A and main relay terminal D. **If there is a common connector:** • Determine the malfunctioning location by inspecting the common connector and terminals for corrosion, damage, or disconnection and by inspecting the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit, or the DC-DC converter (i-ELOOP) terminal 4A-related wiring harness. Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) Go to Step 11.
6	**INSPECT ENGINE2 15 A FUSE** • Remove the ENGINE2 15 A fuse. • Inspect the ENGINE2 15 A fuse. • Is there any malfunction?	Yes	If the fuse is blown: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Main relay terminal C—PCM terminal 2S — Main relay terminal C—PCM terminal 2T **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. — Replace the fuse. If the fuse is damaged: • Replace the fuse. Go to Step 11.
		No	Reinstall the ENGINE2 15 A fuse, then go to the next step.

STEP	INSPECTION		ACTION
7	**INSPECT PCM POWER SUPPLY CIRCUIT FOR SHORT TO GROUND** • Main relay is removed. • Inspect for continuity between main relay terminal C (wiring harness-side) and body ground. • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: — Main relay terminal C—PCM terminal 2S — Main relay terminal C—PCM terminal 2T **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to Step 11.
		No	Go to the next step.
8	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 11.
		No	Go to the next step.
9	**INSPECT PCM GROUND CIRCUIT FOR OPEN CIRCUIT** • Main relay is removed. • Verify that the PCM connector is disconnected. • Inspect for continuity between the following harnesses: — PCM terminal 1BZ (wiring harness-side)—Body ground — PCM terminal 2AA (wiring harness-side)—Body ground • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • PCM terminal 1BZ (wiring harness-side)—Body ground • PCM terminal 2AA (wiring harness-side)—Body ground **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Inspect for the following: — Open circuit between PCM and body ground — Loose or lifting ground point • Repair or replace the malfunctioning part. Go to Step 11.
10	**INSPECT PCM POWER SUPPLY CIRCUIT FOR OPEN CIRCUIT** • Main relay is removed. • Verify that the PCM connector is disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Main relay terminal C—PCM terminal 2S — Main relay terminal C—PCM terminal 2T • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Main relay terminal C—PCM terminal 2S • Main relay terminal C—PCM terminal 2T **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.

STEP	INSPECTION		ACTION
11	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Start the engine and warm it up completely. • Perform the Pending Trouble Code Access Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the PENDING CODE for this DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
12	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.